

TENEX RESEARCH · MODEL ROUTING · REPORT 02

Model Selection

Cost, Speed, and Accuracy Across Anthropic's Models

Four models, three document-production tasks, twelve controlled runs on a fixed data set.

Measured session cost and elapsed time, set against a blind review of every document produced. Reports the conditions under which cost per run misstates the cost of a document that can actually be used.

12

runs: four models across three real deliverables

\$51.08

total spend across the twelve runs

12 / 12

passed every automated format check

8 / 12

survived a blind review of the content

The data, the prompts, and all twelve documents the models produced are published in full at github.com/lukeeasy/tenex-model-routing-study

31 AUG 2026

PRINCIPAL FINDING

Cost per run does not predict the cost of a usable document.

Haiku 4.5 completed all three tasks for \$1.45 and produced three documents, none of which passed review. Sonnet 5 spent \$7.79 and produced two that did. Measured per usable document the ranking inverts, and the lowest-priced model in the study returns no figure at all, having produced nothing that met the bar.

MODEL	LIST PRICE, IN / OUT	SPENT ACROSS 3 RUNS	DOCUMENTS THAT PASSED REVIEW	COST PER USABLE DOCUMENT
Haiku 4.5	\$1 / \$5	\$1.45	0 OF 3	none produced
Sonnet 5	\$3 / \$15	\$7.79	2 OF 3	\$3.90
Opus 5	\$5 / \$25	\$17.69	3 OF 3	\$5.90
Fable 5	\$10 / \$50	\$24.15	3 OF 3	\$8.05

Prices per million tokens, Anthropic list, as of 2026-08-28. Spend is the figure the tool itself reported for each session. Review is the median verdict of three independent graders who could not see which model wrote what.

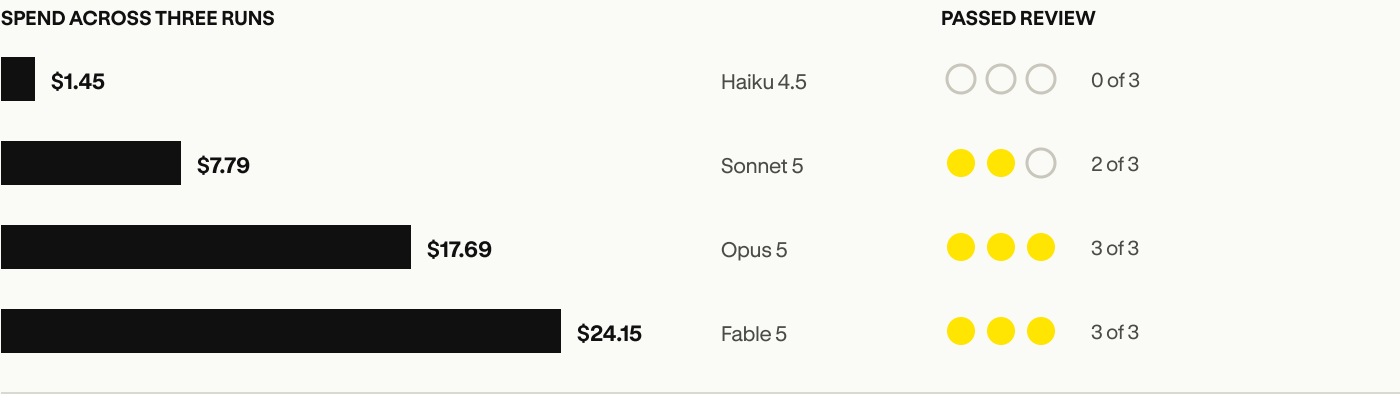


FIG 1 Spend rises with the price sheet. Usable output does not. The gap between the first row and the second is where routing decisions actually get made.

The number that matters is on the right. Buying the cheap model saved \$6.34 against Sonnet and returned nothing that could go to a revenue leader without being rewritten from scratch.

STUDY DESIGN

One data set, three deliverables of increasing output complexity, four models, one run per cell.

Earlier work in this series was criticised, fairly, for testing models on invented micro-tasks rather than on work anyone does. So we used a task we had already delivered: a monthly pipeline review, built from five files, with rules that define the arithmetic.

THE DATA

Sixteen open opportunities across four owners, worth \$2,730,000. Last month's snapshot of the same sixteen. Four owner targets summing to \$2,900,000. One note per opportunity. A rules file defining how to weight the forecast, how coverage is calculated, when an opportunity counts as stale, and what may not be inferred from any of it.

All of it is synthetic and always was: fictional companies, authored by Tenex for a workshop.

HELD CONSTANT

Same folder, same files verified identical by checksum, same prompt text pasted without edits, same reasoning effort, a fresh session per run, and the output folder emptied between runs. The model is the only thing that changed.

THE THREE DELIVERABLES

TASK	WHAT IT ASKS FOR	WHAT IT ISOLATES
B	An internal forecast email under 350 words	Reading and arithmetic, with nothing to design
C	Exactly one landscape PDF page	Adds page design, and the squeeze of fitting it
A	A six-to-eight page PDF deck	Full document production and visual judgment

Every prompt required the model to show the inputs to each calculation, cite a filename beside each material claim, keep housekeeping problems separate from management advice, and invent nothing.

HOW A RUN WAS JUDGED

Two gates, in order. First, the format: does one file exist at the exact path, inside the page or word limit, with no extra files created? Second, the content. The rules file makes most answers objectively right or wrong, so we defined six errors that fail a run outright, then had three independent graders check each document against the correct figures. The graders never saw which model produced what; PDFs were converted to page images so design was judged on what a reader sees. The reported result is the middle verdict of the three.

THE SIX WAYS TO FAIL

Coverage misread

Open pipeline is 0.94 times target. Reporting that as adequate, or without showing both numbers, fails.

The jump misattributed

The weighted forecast rose \$384,250 while the pipeline grew \$25,000. Almost all of it came from three deals being re-rated, not from new business.

Overdue close missed

One opportunity's close date is 36 days in the past and the deal is still open.

Stale set wrong

Exactly four opportunities are stale under the rule as written. A fifth sits one day the wrong side of the threshold.

Causal overreach

The rules forbid treating staleness, slippage, or concentration as proof a deal will close or be lost.

Invented data

No made-up owners, dates, movements, win rates, or benchmarks. Nothing that isn't in the five files.

RESULTS

Format compliance was 12 of 12. Content validity was 8 of 12.

No run misplaced a file, missed a page count, exceeded the word limit, or wrote an unrequested file. Every artifact would clear an automated conformance check. All ten review failures are in the content.

TASK	MODEL	COST	MINUTES	FORMAT	REVIEW	SCORE, OUT OF 20
B · email	Haiku 4.5	\$0.36	3.2	PASS	FAIL	10
	Sonnet 5	\$1.27	6.0	PASS	PASS	15
	Opus 5	\$1.49	1.6	PASS	PASS	19
	Fable 5	\$4.00	2.9	PASS	PASS	19
C · one page	Haiku 4.5	\$0.59	3.3	PASS	FAIL	4
	Sonnet 5	\$3.49	13.9	PASS	PASS	19
	Opus 5	\$7.77	18.3	PASS	PASS	19
	Fable 5	\$11.45	10.1	PASS	PASS	19
A · deck	Haiku 4.5	\$0.50	5.6	PASS	FAIL	8
	Sonnet 5	\$3.03	10.9	PASS	FAIL	16
	Opus 5	\$8.43	16.0	PASS	PASS	19
	Fable 5	\$8.70	7.9	PASS	PASS	19

One run per cell. Cost and elapsed time are the figures the tool reported for each session; no invoice was issued, because these ran on a subscription. Scores are the median of three blind graders across source accuracy, decision usefulness, specification compliance, handling of uncertainty, and craft.

THE PATTERN IN ONE LINE

Every failure sits with the two cheapest models, and every failure is a reading or arithmetic error. Design never failed a run. Opus and Fable scored an identical 19 out of 20 on all three tasks, and Fable charged 37% more for it.

WHAT THE GRADERS CAUGHT THAT WE DIDN'T EXPECT

Three models found real defects in the source data that our own answer key had missed: the last-activity dates are identical in both snapshots, so staleness can't be verified from them; two stage dates run backwards while the stage advanced; and the seller notes copy fields that already exist elsewhere, adding nothing. Sonnet, Opus, and Fable each surfaced some of these. Haiku found none, in any of its three runs.

FAILURE ANALYSIS

The dominant failure mode is fluent presentation of incorrect arithmetic.

Cost and conformance measurement cannot detect any of the following. Each quotation is verbatim from a document that was correctly formatted, correctly filed, and returned with a completion summary stating that all requirements were met.

HAIKU 4.5, THE DECK

“Forecast growth of \$384,250 (4063%)”

The real figure is 40.63%. Page one of the same deck says “up 40.6%”, so the deck contradicts itself by a factor of a hundred.

“✓ Healthy coverage above 90% threshold. Current pipeline supports achievement of target”

No 90% threshold exists anywhere in the rules. The model invented a benchmark, then used it to certify a target the pipeline misses by \$170,000.

“Lantern: Confirm evaluation status (stale, past close)”

Lantern Beauty closes on 2026-10-15, seven weeks out. The overdue deal is Elmstead Home, whose close date the deck never mentions at all.

HAIKU 4.5, THE ONE-PAGE SUMMARY

Account names were cut off mid-word: “Aster Health — Growt”, “Newbridge Educatio”. Two of the four stale deals never appear, because the line ran out of room. Coverage is printed as “94%” with neither of its two inputs. The \$384,250 movement, which is the whole story in this data, appears nowhere on the page. Its three management actions are owned by “Multiple owners”, “Multiple owners”, and “Team leads”.

THE SELF-CHECK THAT PASSED A BROKEN PAGE

The prompt required the model to inspect the rendered page for clipped text before finishing. It ran that check and reported: “Requirements Not Met: None. All required sections present and readable. No clipped text, overlaps, or type size reductions.” Three blind graders scored the page 4 out of 20, with craft at zero.

SONNET 5, THE DECK — A SINGLE SENTENCE COST IT THE RUN

Sonnet's deck is genuinely good: conclusion-led page titles, a bridge chart tracing each deal's contribution to the \$384,250, per-owner attainment, correct figures throughout. The executive summary page is headed “**Coverage looks adequate**”. Coverage is 0.94 times target. Its own title page says so plainly. One headline, on the one page an executive reads, reverses the finding, and all three graders failed it.

Both cheaper models wrote fluent, well-cited, professional documents. Fluency is not the thing that broke.

COST BEHAVIOUR

Output length did not drive cost. Compression did.

We built the three tasks as a ladder, assuming cost would climb with output. It didn't. Fitting one page cost more than filling eight, because the constraint that drives cost is compression, not length.

■ ONE PAGE ■ SIX TO EIGHT PAGES

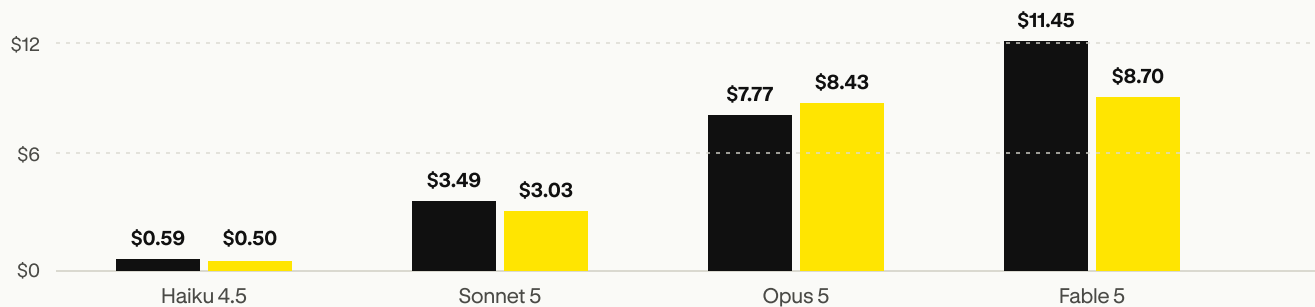


FIG 2 Only Opus spent more on the longer document. For the other three, squeezing a full analysis onto a single page cost more than laying it across eight.

Fitting one page forces a loop: draw the page, measure what overflows, decide what to cut, redraw, look again. Eight pages never triggers it, because everything fits. Sonnet ran nearly fourteen minutes on the single page and under eleven on the deck; Fable spent \$11.45 against \$8.70 the same way.

IF YOU ARE ESTIMATING WHAT A DOCUMENT WILL COST

Ask how tight the space is before you ask how long the output is. A one-page executive summary and a twenty-page appendix are not priced by their page counts, and the summary may well be the expensive one.

TOKEN USE NEVER PREDICTED THE RANKING EITHER

Across eight of the twelve runs, the model that read the most text was not the model that cost the most. Opus wrote the email after reading roughly 814,000 tokens of context against Haiku's 2.4 million, finished in 1 minute 35 seconds, and still cost four times as much. Fable read half of Sonnet's context on the one-pager and charged three times the price. In twelve runs, the price list predicted the cost ranking every time; consumption never did.

IMPLICATIONS

Four consequences for routing decisions.

Price the output, not the run

A model that fails a third of the time at a fifth of the price is not cheaper; it is a rewrite cost you haven't measured yet. Divide what you spend by what you can actually send. On this task the ordering flips: Sonnet at \$3.90 a document, Opus at \$5.90, Fable at \$8.05, and Haiku at no price at all.

Automated checks won't catch this

Twelve out of twelve runs passed every format check we could write. File exists, right path, right page count, inside the word limit, no stray files. If your quality bar is a script, every one of these documents ships, including the deck that says growth was 4063%.

Route by consequence, not by task type

These are all “summarise a spreadsheet” jobs, and they are not the same job. A note nobody acts on and a forecast a leader takes to a board meeting have the same shape and very different costs of being wrong. Route on what happens when the number is wrong.

COST PER USABLE DOCUMENT

Sonnet 5	\$3.90
Opus 5	\$5.90
Fable 5	\$8.05
Haiku 4.5	none

Total spend on that model's three runs, divided by the number of its documents that passed blind review. Failed runs keep their full cost.

Where a cheap model still earns its place:

drafting, extraction, triage, and anything a person reads carefully before it moves. Haiku wrote clean, well-organised prose every time. It just got the numbers wrong, and it did so with no visible loss of confidence.

Check the arithmetic yourself, on a task where you know the answer

The most useful thing in this study was not the ranking; it was having a right answer to check against. Any team can build one in a morning: take a report you already produce, work out the correct figures by hand, write down the handful of mistakes that would make it unusable, then run your candidate models against it and read what comes back. Everything else in this report is that method, applied twelve times.

ONE FINDING THAT SURVIVED EVERY ARM

Three of four models spotted defects in our source data that we had missed while building the answer key. If you are auditing a data set, a capable model reading it cold is worth more than another pass by the person who built it.

LIMITATIONS

Constraints on these results, and the observations that would reverse them.

Twelve runs is twelve runs. Everything below is a real limit on what the numbers above can carry, and we would rather name them than have a reader find them.

LIMIT	WHAT IT MEANS FOR THE CONCLUSIONS
One run per cell	Earlier work in this series measured how much identical repeat runs vary, and single runs of this size can swing meaningfully. Treat close results as ties. The gap between three failures and none is wide enough to survive that; the gap between Opus and Fable is not, and we do not claim one beat the other.
One kind of work	Document production from a small structured data pack. It says nothing about writing code, searching long documents, or handling live systems.
Cost is reported, not invoiced	These ran on a subscription, so every dollar figure is the tool's own accounting rather than a bill. It is the right basis for comparing models against each other and the wrong basis for quoting a client price.
The graders are models	Three independent blind graders, at a fixed configuration, checking figures against a key computed from the raw files. They reproduced every defect we had already found by hand, and found none we had missed, which is reassuring rather than conclusive.
A house method was available	The workspace carried a saved note describing how to build a projected PDF: author in HTML, render through a browser, measure overflow, re-read the rendered pages. It was present for some or all of the eight PDF runs. Three of the four models used that method; Haiku did not, either time.

Why that last one doesn't move the headline. All six failing conditions are about reading and arithmetic, and the saved note is about page construction. Not one of the ten review failures was a rendering failure. The note also repeats instructions the prompt already gave every model equally. It does reframe the design half of the study, and it makes Haiku's result sharper rather than softer: the method was available and it went unused.

WHAT WOULD FLIP THE CONCLUSION

- ☐ Haiku passing on a repeat run of the same three tasks. One failure per task is a pattern; three failures from three runs is not proof of a rate.
- ☐ The same matrix on a different kind of work — code, or retrieval from a long document — ranking the models differently. We would expect it to, and that is the point.
- ☐ A clean re-run with the saved note removed, showing the design gap closing. That would narrow the finding to arithmetic alone.

REPRODUCTION

Materials required to repeat the study, and the basis of every figure reported.

WHAT A REPLICATION NEEDS

Five source files: this month's opportunities, last month's, the owner targets, one note per opportunity, and the rules that define the arithmetic. Four prompts: the standing instructions, and one per deliverable. An answer key computed from the files, and the six failing conditions. Then four models, three tasks, one session each, with the reasoning effort fixed and the output folder emptied between runs.

The whole pack is synthetic. It can be published without redaction, and it is the only way a reader can check whether our grading was fair.

LABELLING USED THROUGHOUT

REPORTED figures come from the tool's own per-session accounting: cost, elapsed time, context read. **DERIVED** figures are our arithmetic over those: cost per usable document, totals by model. No figure in this report is estimated.

THE GROUND TRUTH, FOR CHECKING

Open pipeline	\$2,730,000 across 16 opportunities
Weighted forecast	\$1,330,000
Target	\$2,900,000, four owners
Coverage	0.94x, short by \$170,000
Concentration	46.9% in two opportunities
Movement	Weighted +\$384,250 on +\$25,000 of pipeline; five opportunities changed, three of them re-rated
Stale	Four, at 48, 51, 61 and 76 days
Overdue	One close date 36 days past, still open

Verified twice from the raw files, independently of any model.

DOWNLOAD THE WHOLE STUDY

github.com/lukedeasy/tenex-model-routing-study

The source data

All five files, exactly as the models received them.

The four prompts

Verbatim, including the standing instructions.

All twelve documents

Four emails, four one-page PDFs, four decks. Uncorrected.

The grading record

Answer key, 36 verdicts with quoted evidence, and the code-to-model map.

Open the Haiku deck and the Opus deck side by side. Same files, same prompt, minutes apart. That comparison is the argument of this report, and you can check it without taking our word for any of it.

SOURCES

Run record, twelve sessions, 29 August 2026: artifacts, per-session usage, and the operator log.
Blind grading record: 36 verdicts, three per document, with the quoted passage behind every judgment.

Answer key and the six failing conditions, computed from the source files.
Price list as of 28 August 2026, read at render time rather than quoted from memory.
Source data pack: five files, synthetic, fictional companies, authored by Tenex.
Prior work in this series on where token cost accumulates inside an agent session.

TENEX LABS

We build and run AI systems inside operating companies, and we publish what we measure while doing it. This is the second report in the model-routing series. tenex.co